Akhil Kothapalli

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Education

Georgia Institute of Technology — Atlanta, GA
Bachelors – Concentration in Networks & Machine Learning; GPA: 3.9 — *Jun 2022 - May 2026*

Experience

Amazon Web Services — Seattle, WA
Software Development Engineer I – Nitro Networking Performance — *June 2026 – Present*

- Develop performance testing infrastructure for AWS Nitro NX cards on the Networking Performance team, identifying customer-impacting bottlenecks and driving improvements across new-generation hardware.
- Optimize network data-path performance by offloading compute-intensive operations from host CPU to dedicated on-card hardware, improving throughput and reducing latency at scale.
- Build and maintain tooling used org-wide to benchmark and validate networking performance across EC2 instance families and hardware generations.

Software Development & Networking Intern — *Jun 2025 – Aug 2025*

- Architected a UDP packet benchmarking suite using Cisco T-Rex for AWS Nitro NX cards, resolving Graviton architectural incompatibilities and replacing unrealistic max-throughput tests with configurable IMIX workloads across 14 real-world scenarios.
- Built a Python Behave-based orchestrator adapting test execution across EC2 instances ranging from 1–96 vCPUs and 1–4 network adapters, with reproducible NixOS images via EC2 Image Builder cutting provisioning from 13 minutes to under 1 minute.
- Validated 200+ Gbps throughput at 55M+ packets/sec on flagship EC2 instances; results contributed to publicly released AWS networking performance data, with the framework adopted org-wide across all AWS networking teams.

Georgia Tech – IFL Lab — Atlanta, GA
Research Assistant – Autonomous Racing Vehicles — *Jan 2024 – Dec 2025*

- Developed autonomous racing algorithms on NVIDIA Jetson (Linux, ROS 2) leveraging LiDAR and computer vision for perception and control.
- Implemented SLAM Toolbox for map generation and AMCL for localization; integrated dynamic Pure Pursuit for path planning and waypoint navigation.
- Boosted racing performance by 23% over prior years by optimizing mapping, planning, and control while building a modular, extensible codebase for future research.

Projects

Transforming Robots Tycoon | *Lua, JSON, Python* — 2023 – Present

- Led creation of multiplayer game based on Transformers attracting over 600,000 players.
- Visualized analytics using Python API to understand how users experience the game.

Real-time Camera Drone | *C, Embedded Systems, WebSockets* — May 2025 – Present

- Engineered camera drone inspired by Rainbow Six Siege with low-latency video streaming and motor control.
- Integrated Flask-SocketIO with Raspberry Pi GPIO for real-time communication.
- Built server with Flask on AWS, leveraging its authentication system to enable secure, cross-device bot control.

Website Bot Protection | *Node.js, Express, JavaScript, CryptoES* — Jan 2025

- Built a multi-stage anti-bot challenge system using behavioral sensor analysis and a semi-prime factorization puzzle to verify human users.
- Implemented AES-encrypted challenge responses, rate limiting, cookie-based session management, and client-side obfuscation to resist automation.

Technical Skills

Coding Languages: Python, C/C++, Lua
Frameworks: Flask, OpenCV, scikit-learn, DPDK, numpy
Developer Tools: Git, Docker, ROS, VS Code, AWS, Linode
AI Tools: Claude Code, Amazon Q, Cursor, Agent Spaces
Operating Systems: Windows, macOS, Linux (Ubuntu, Raspberry Pi OS, Amazon Linux, CentOS)